

DaSEH Day 8 Cheatsheet

Statistics

Functions

Library/Package	Piece of code	Example of usage	What it does
Base R corrplot	<code>cor()</code> <code>corrplot()</code>	<code>cor(x, y)</code> <code>corrplot(cor_mat, type = "upper", order = "hclust")</code>	Calculates correlation between two vectors. Creates a correlation matrix plot.
Base R	<code>t.test()</code>	<code>t.test(x, y, alternative = "two.sided")</code>	Performs one- and two-sided t-tests.
broom	<code>tidy()</code>	<code>tidy(t_test_result)</code>	Converts statistical objects to tidy data frames.
Base R	<code>wilcox.test()</code>	<code>wilcox.test(x, y)</code>	Performs nonparametric Wilcoxon signed-rank or rank-sum tests.
Base R	<code>shapiro.test()</code>	<code>shapiro.test(x)</code>	Tests normality with the Shapiro-Wilk test.
Base R	<code>ks.test()</code>	<code>ks.test(x)</code>	Tests a distribution with the Kolmogorov-Smirnov test.
Base R	<code>var.test()</code>	<code>var.test(x, y)</code>	Compares two variances with Fisher's F-test.
Base R	<code>chisq.test()</code>	<code>chisq.test(x, y)</code>	Performs chi-squared contingency table and goodness-of-fit tests.
Base R	<code>summary()</code>	<code>summary(linear_model_result)</code>	Summarizes an object, including models and statistical test results.
Base R	<code>glm()</code>	<code>glm(x ~ y, data = df, family = binomial())</code>	Fits generalized linear models; specify distribution and link with <code>family</code> .
epitools	<code>oddsratio()</code>	<code>oddsratio(x, y)</code>	Calculates odds ratios.

Data Output

Functions

Library/Package	Piece of code	Example of usage	What it does
tidyverse (readr)	<code>write_csv()</code>	<code>write_csv(df, "file.csv")</code>	Writes a data frame to a CSV file.
tidyverse (readr)	<code>write_delim()</code>	<code>write_delim(df, "file.txt", delim = ":")</code>	Writes a data frame to a delimited file.
tidyverse (readr)	<code>write_rds()</code>	<code>write_rds(df, "file.rds")</code>	Saves a single R object to an RDS file.
tidyverse (readr)	<code>read_rds()</code>	<code>df <- read_rds("file.rds")</code>	Reads a single R object from an RDS file.
Base R	<code>save()</code>	<code>save(df1, df2, file = "file.RData")</code>	Saves one or more R objects to an RData file.
Base R	<code>load()</code>	<code>load("file.RData")</code>	Loads R objects from an RData file into the environment.

* This cheatsheet format was adapted from Alex's Lemonade Stand materials ([source](#)).

* Find more resources at <https://daseh.org>.